LIQUID LINE (Continued)

(2) Remove the tape or plugs from the liquid line fitting and the condenser port.

(3) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the liquid line fitting. Use only the specified O-rings as they are made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the compressor in the vehicle.

(4) Connect the liquid line to the A/C condenser.

(5) Install the nut that secures the liquid line to the A/C condenser. Tighten the nut to 12 N·m (105 in. lbs.).

(6) Install the liquid line into the plastic retaining clips located on the right inner fender.

(7) Remove the tape or plugs from the liquid line fitting and the evaporator port.

(8) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the liquid line quick-connect fitting. Use only the specified O-rings as they are made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the compressor in the vehicle.

(9) Connect the liquid line to the A/C evaporator (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/REFRIGERANT LINE COUPLER - INSTALLATION).

(10) Install the secondary retaining clip onto the spring-lock coupler that secures the liquid line to the A/C evaporator.

(11) On LHD models, go to Step 17. On RHD models, proceed to Step 12.

(12) On RHD models, remove the tape or plugs from the two liquid line fittings.

(13) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the liquid line quick-connect fitting. Use only the specified O-rings as they are made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the compressor in the vehicle.

(14) Connect the rear section of the liquid line to the front section of the liquid line (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/REFRIGERANT LINE COUPLER - INSTALLATION).

(15) Install the secondary retaining clip onto the spring-lock coupler that secures the rear section of the liquid line to the front section of the liquid line.

(16) Install the rear section of the liquid line into the retaining clips located on the dash panel.

(17) If removed, install the engine air cleaner housing (Refer to 9 - ENGINE/AIR INTAKE SYSTEM/AIR CLEANER HOUSING - INSTALLATION).

(18) Reconnect the negative battery cable.

(19) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(20) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

REFRIGERANT

DESCRIPTION

The refrigerant used in this air conditioning system is a HydroFluoroCarbon (HFC), type R-134a. Unlike R-12, which is a ChloroFluoroCarbon (CFC), R-134a refrigerant does not contain ozone-depleting chlorine. R-134a refrigerant is a non-toxic, non-flammable, clear, and colorless liquefied gas.

Even though R-134a does not contain chlorine, it must be reclaimed and recycled just like CFC-type refrigerants. This is because R-134a is a greenhouse gas and can contribute to global warming.

OPERATION

R-134a refrigerant is not compatible with R-12 refrigerant in an air conditioning system. Even a small amount of R-12 added to an R-134a refrigerant system will cause compressor failure, refrigerant oil sludge or poor air conditioning system performance. In addition, the PolyAlkylene Glycol (PAG) synthetic refrigerant oils used in an R-134a refrigerant system are not compatible with the mineral-based refrigerant oils used in an R-12 refrigerant system.

R-134a refrigerant system service ports, service tool couplers and refrigerant dispensing bottles have all been designed with unique fittings to ensure that an R-134a refrigerant system is not accidentally contaminated with the wrong refrigerant (R-12). There are also labels posted in the engine compartment of the vehicle and on the A/C compressor to identify that the A/C system is equipped with R-134a refrigerant.

REFRIGERANT LINE COUPLER

DESCRIPTION

Spring-lock type refrigerant line couplers are used to connect the A/C liquid line to the A/C evaporator and, on RHD models the front section of the liquid line to the rear section of the liquid line. Secondary retaining clips are installed over the connected couplers for added protection.

The spring-lock refrigerant line couplers require special disconnect tools for disengaging the two coupler halves.